Hamoud Alshammari

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RESEARCH INTERESTS

Estimation and uncertainty quantification for space systems: orbit determination and maneuver detection, GNSS/PNT integrity and LEO navigation, satellite geodesy and reference frames, and open, well-tested software for astrodynamics.

EDUCATION

The University of Texas at Austin, Cockrell School of Engineering — Austin, TX, USA
M.S.E. in Aerospace Engineering · GPA 3.9/4.0 · Academic advisor: Prof. Ryan P. Russell — *Jan 2023 – Dec 2024*
Coursework: Orbit Determination; Satellite Geodesy; Stochastic Estimation & Control; Advanced Estimation Theory; Satellite Control Systems; Optimal Control; Celestial Mechanics; SAR; LiDAR; LEO Satellites for EO; ML in the Geosciences.

University of Wisconsin–Madison, College of Engineering — Madison, WI, USA
B.S. in Engineering Mechanics (Aerospace Engineering focus) — *Jan 2018 – Dec 2020*

PROFESSIONAL EXPERIENCE

Neo Space Group — Riyadh, Saudi Arabia
Aerospace Technology Engineer — *May 2025 – Present*

- R&D and technical due diligence across satellite communications, PNT, Earth observation and NB-IoT; independent quantitative checks of vendor performance claims and proofs of concept.
- Constellation-geometry analysis, including dilution-of-precision studies comparing single-shell and multi-inclination LEO designs under realistic elevation masks; link budgets and coverage/capacity models.
- Briefings for executive leadership and partners; defined proof-of-concept scope, milestones and success metrics.

RESEARCH EXPERIENCE

Center for Space Research, The University of Texas at Austin — Austin, TX, USA
Research Engineering/Scientist Professional — *Jan 2025 – Apr 2025*
Graduate Research Assistant — *Feb 2024 – Dec 2024*

- Lunar geodesy and reference-frame research in a collaboration between CSR and CU Boulder's Colorado Center for Astrodynamics Research (Profs. S. Bettadpur, L. Magruder; Prof. R. S. Nerem at CU).
- Reviewed lunar frame realizations and frame ties; assessed multi-source and NASA datasets for a consistent lunar frame.
- Co-authored internal technical documentation and presentations recording assumptions and frame-tie strategies for team reuse.

SELECTED GRADUATE PROJECTS

- **Orbit determination.** EKF/UKF on simulated range/range-rate data from three ground stations (6-day arc + 1-day prediction); J2 dynamics, Monte Carlo process-noise tuning, residual and RIC covariance-realism analysis.
- **Attitude determination & control.** Closed-loop ADCS with environmental torques, reaction wheels, PD Earth-pointing and a six-state MEKF for attitude and gyro bias; 100-run Monte Carlo validation.
- **Orbit maintenance.** Replicated TerraSAR-X's 11-day sun-synchronous repeat orbit with J2 and drag; planned ΔV maneuvers within a 250 m control tube.
- **Geodesy & remote sensing.** Degree-variance spectra of gravity-field functionals from spherical-harmonic models; time-domain SAR back-projection; CNN tree detection from satellite/LiDAR data vs. DeepForest.

OPEN-SOURCE SOFTWARE (github.com/OrbitAR7)

- **GNSS & navigation:** *gnss-integrity-raim* (Python; chi-square FDE, protection levels validated by Monte Carlo), *gnss-pseudorange-solver* (C++; least squares, Sagnac, DOP), *constellation-dop-analyzer* (Python; DOP from real GPS/Starlink TLEs).
- **Estimation & astrodynamics:** *EstimationToolkit.jl* (Julia; 36 filters incl. IMM/GPB and particle flow, 568 tests), *tle-sgp4-cpp* (C++; SGP4 matching the reference to <1 µm), *orbit-propagator-cpp* (C++; two-body + J2, validated).
- **Simulation & geodesy:** *basilisk-link-budget* (Python; Basilisk tests), *gibbs-phenomenon-visualizer* (Python; SH leakage).

SKILLS

Programming: Python (NumPy, SciPy, PyTorch), MATLAB & Simulink, Julia, C/C++, shell scripting for HPC, Git, LaTeX.
Methods: EKF/UKF/MEKF, particle filters, batch least squares, Monte Carlo, SGP4/J2/drag propagation, spherical harmonics, SAR.
Languages: Arabic (native), English (fluent). **Service:** Habitat for Humanity builds (Houston, Raleigh, Wisconsin), 2018–20.

REFERENCES

Prof. Lori Magruder — Director, CSR, UT Austin
lori.magruder@austin.utexas.edu

Prof. Srinivas Bettadpur — Aerospace Eng. & CSR, UT Austin
srinivas@csr.utexas.edu

Prof. Moriba Jah — Aerospace Eng. & Oden Inst., UT Austin
moriba@utexas.edu

Prof. Ryan P. Russell — Aerospace Engineering, UT Austin
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